Causal Inference for Asset Pricing

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This paper: A framework for causal inference with asset prices and quantities

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  - Index inclusions, mutual fund reclassifications, ...
  - Fed randomly purchases some bonds but not others, broad QE announcements, operation twist

- **Market macrostructure questions** (Haddad Muir 2025)
  - Quantitative easing, international capital flows
  - ESG mandates, rise of passive investing, ...

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Economic Framework

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Canonical causal inference
Angrist Pischke 2009

↓

Economic Framework

↓

Demand systems
Koijen Yogo 2019, Gabaix Koijen 2023
Haddad He Loualiche 2025

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A Framework?

Wishlist:

- **Flexible but parsimonious**: capture the forces of many demand structures, particularly specific to finance (*"everything is connected: portfolios and prices"*)
  - Key missing piece of existing models: elasticity to price of factors/characteristics = substitution depends on characteristics
- **Simple portable assumption**: can discuss plausibility without fully taking a stand on microfoundation
  - Counterpart of canonical causal inference assumption (e.g. Angrist Pischke 2009)
  - Systematic set of diagnostics, empirical design, ...
- **Easy transparent estimation**: set of IV/diff-in-diff regressions
  - "Separable": map from different types of natural experiment to different counterfactuals
  - Precisely define what is a valid instrument
  - Lots of work on finding instruments, not the focus here

Claim: a single assumption *"homogeneous substitution conditional on observables"* provides all this

Related literature

- **Asset pricing using causal inference methods**
  - Shleifer (1986); Harris, Gurel (1986); Chang, Hong, Liskovich (2014); Pavlova, Sikorskaya (2023); Greenwood, Sammon (2024); Gompers, Metrick (2001); Coval, Stafford (2007); Lou (2012); Ben-David, Li, Rossi, Song (2022); Hartzmark, Solomon (2022); Krishnamurthy, Vissing-Jorgensen (2011); Haddad, Moreira, Muir (2021, 2025); Selgrad (2024); Du, Tepper, Verdelhan (2018); Greenwood, Vissing-Jorgensen (2018); Haddad, Muir (2021); Chen, Chen, He, Liu, Xie (2023); ...
- **Structural approach and demand systems**
  - Koijen, Yogo (2019, 2024); Koijen, Richmond, Yogo (2024); Haddad, Huebner, Loualiche (2024); van der Beck (2024); Lu, Wu (2023); Gabaix, Koijen (2024); Bretscher, Schmid, Sen, Sharma (2024); Jansen, Li, Schmid (2024); Fang (2023); Fang, Xiao (2024); Graves (2025); ...
  - Li, Lin (2024); Chaudhary, Fu, Li (2023); Aghaee (2024); An, Huber (2025); Peng, Wang (2023); Fuchs, Fukuda, Neuhann (2024); ...
- **What we don't do:**
  - Strategic responses (Haddad, Huebner, Loualiche, 2024)
  - Dynamics (Greenwood, Hanson, Liao, 2018; Huebner, 2024; Gabaix, Koijen, 2024; He, Kondor, Li)
  - State-contingent shocks (Haddad, Moreira, Muir, 2025)
- **Spillovers/substitution outside asset pricing:**
  - Berry, Levinsohn, Pakes (1995), Berg, Reisinger, Streitz (2021); Chodorow-Reich, Nenov, Simsek (2021); Guren, McKay, Nakamura, Steinsson (2021), Huber (2023); Wolf (2023); ...

The Problem

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1. Microeconomics reminder
2. Asset pricing reminder

Question

How do an investor's portfolio decisions respond to prices?

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$N+1$ assets, asset 0 is the numeraire. Payoffs $(\Pi_0, \ldots, \Pi_N)$ with distribution F. Prices $P = (P_1, \ldots, P_N)$.

$$\max_{Q_0, Q_1, \ldots, Q_N} \mathbb{E}\left[u\left(\sum_{i=0}^{N} Q_i \Pi_i\right)\right]$$

$$\text{s.t.} \quad Q_0 + \sum_{i=1}^{N} Q_i P_i \leqslant \bar{Q}_0 + \sum_{i=1}^{N} \bar{Q}_i P_i$$

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$$\Longrightarrow \quad \text{Demand function} \quad (Q_1^*, \ldots, Q_N^*) = D(P, F, \text{preferences})$$

- *Structural* (Hurwicz 1966): stable independently of the other players in the equilibrium environment
- Rationality, utility maximization, purely pecuniary preferences not necessary for demand function (Becker 1962)

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How do an investor's portfolio decisions respond to prices?

$$\underbrace{\mathcal{E}}_{N \times N} = \frac{\partial D}{\partial P} = \left[\frac{\partial D_i}{\partial P_j} \right]_{ij}$$

Elasticity matrix: sensitivity of demand to prices

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- Mean-variance: $D = \frac{1}{\gamma}\Sigma^{-1}(M - P)$, $\mathcal{E} = -\frac{1}{\gamma}\Sigma^{-1}$
- Could be log, levels, shares, changes or not, ...
- Crucial for macrostructure questions: e.g. price impact $\mathcal{E}_{agg}^{-1}$, how do shifts in demand affect prices?

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- *How does CalPERS adjust its position in 10-year corporate bonds of Ford and GM when their spread changes?*
- *How does CalPERS adjust relative positions of long and short term bonds when the term premium changes?*

An Example: CalPERS and Corporate Bonds

- Prices have moved **and no other news**. CalPERS adjusts its bond portfolio:

	Price change	Change in position
1. 10-yr Ford	+ 5%	sell 200
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⋮	⋮	⋮

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$$\Delta D_1 = \underbrace{\mathcal{E}_{11}\Delta P_1}_{\text{became more expensive}} + \underbrace{\mathcal{E}_{12}\Delta P_2}_{\text{substitutes from GM}} + \underbrace{\sum_{k\geqslant 3}\mathcal{E}_{1k}\Delta P_k}_{\text{substitutes from First Solar, ...}}$$

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→ **Stuck without additional assumptions** (Deaton Muellbauer 1980)

Learning from Standard Finance Models

APT-style model: factor model with i.i.d. residuals, $\alpha_i \perp \beta_i$:

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- *Force 1:* factor management, if expected return only depends on exposures β
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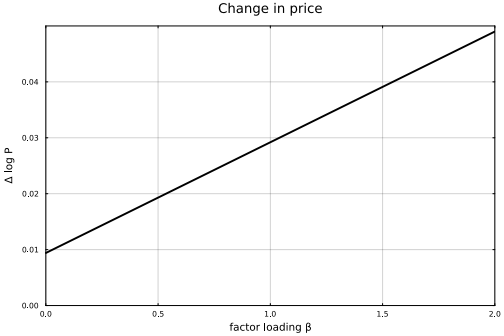
Elasticity: changing prices = changing expected returns

$$\mathcal{E} = \underbrace{-\frac{1}{\gamma\sigma_\epsilon^2}\mathbf{I}}_{\text{diagonal}} + \underbrace{\boldsymbol{\beta}\Psi\boldsymbol{\beta}'}_{\text{substitution matrix}}$$

Why is Everything Needed?

There was a shock to the supply of various assets and you see the price of high-beta assets increase more than those of low-beta assets. What was the supply shock?



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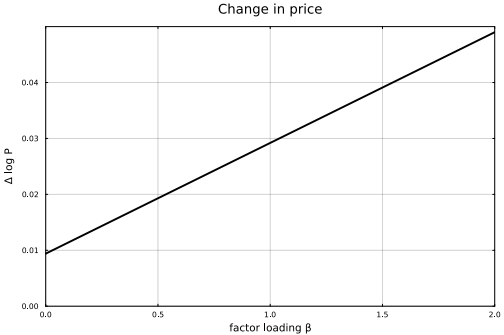
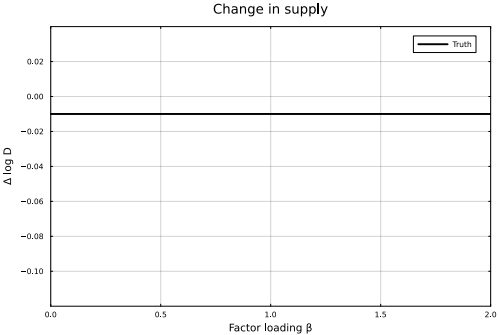
  - factor supply decreased → lower risk premium → price of high β assets increase more





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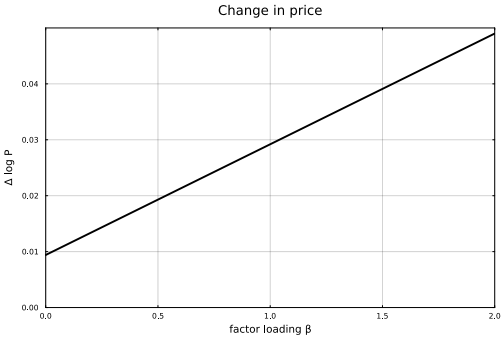

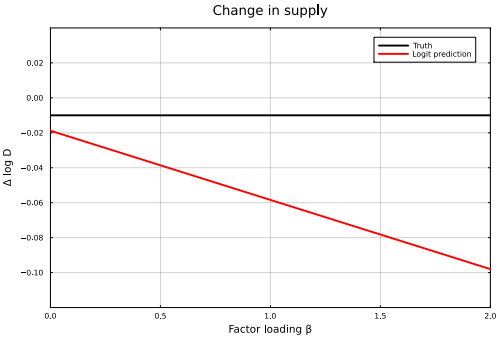

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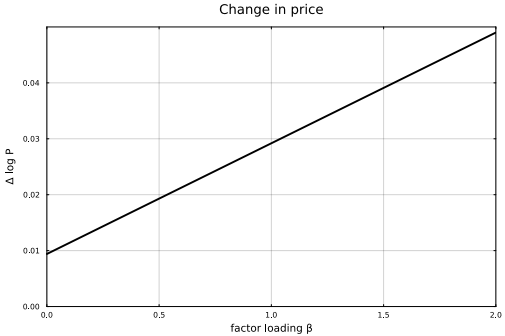
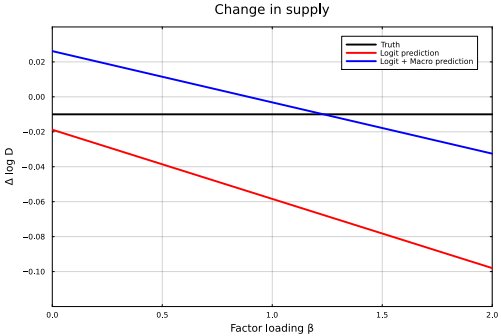
  - Add a macro elasticity (Gabaix Koijen 2023): does not fix it





In The Real World

- **QE across maturities** (e.g. Krishnamurthy Vissing-Jorgensen 2011, Haddad Moreira Muir 2024), see also effect of Treasury supply (Greenwood Vayanos 2014)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	1M	3M	6M	1Y	2Y	3Y	5Y	7Y	10Y	20Y	30Y
QE Event	-0.10	-0.06	-0.32**	-0.37**	-0.51**	-0.58**	-1.09***	-1.34***	**-1.39*** **	-0.90***	-0.76***
	(0.26)	(0.19)	(0.16)	(0.17)	(0.20)	(0.22)	(0.25)	(0.25)	(0.24)	(0.24)	(0.23)

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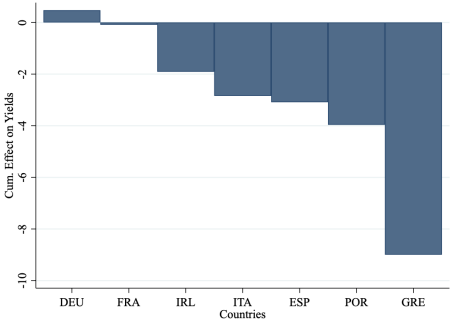
- **EU asset purchases across countries** (e.g. Haddad Moreira Muir 2025)



Two Paths Forward

- **Fully specified model:** choose a microfoundation and estimate the corresponding model
  - Estimate covariance matrix, then risk aversion
  - Impose non-linear restriction on factor model and obtain logit (Koijen Yogo 2019)

- **Causal-inference-style:** impose elementary restriction keeping as much flexibility on mechanism as possible while letting the data speak
  - Useful if skeptical of classical risk motive, rationality, and absence of frictions
  - Canonical assumption (SUTVA): when I give you medication, it affects your health but not the control group's health
  - Does not work here: demand for each asset only depends on its own price $D_i(P_i) \Rightarrow$ diagonal $\mathcal{E}$

Next: a causal inference framework with substitution/spillovers suited to portfolio choice

Framework

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A simple assumption:

- **Homogeneous substitution conditional on observables**
  - CalPERS substitutes across bonds based on their observables (e.g. duration, greenness) only

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$$\textit{GM:} \quad \Delta D_2 = \mathcal{E}_{22}\Delta P_2 + \mathcal{E}_{21}\Delta P_1 + \sum_{k\geqslant 3} \underbrace{\mathcal{E}_{2k}}_{=\mathcal{E}_{1k}} \Delta P_k$$

- **Compare bonds with same observables:** Ford vs. GM
  - E.g.: CalPERS adjusts Ford and GM equally in response to price of First Solar $\mathcal{E}_{13} = \mathcal{E}_{23}$

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$$\textit{Diff-in-diff:} \quad \Delta D_1 - \Delta D_2 = \widehat{\mathcal{E}}(\Delta P_1 - \Delta P_2) \text{ if same relative elasticity}$$

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$\rightarrow$ *comparing assets with same observables differences out substitution*

Formal Setup

- **Homogeneous substitution conditional on observables X**

$$\boxed{\mathcal{E}_{il} = \mathcal{E}_{jl} \text{ if } X_i = X_j} \quad \text{for all } i, j \in \mathcal{S}, \text{ and } l \neq i, j,$$

  - If price of a 3rd asset moves, response of demand for 2 assets with same observables is the same
  - Parametrize linearly: $\mathcal{E}_{il} = \mathcal{E}_{\text{cross}}(X_i, X_l) = X_i' \boldsymbol{\mathcal{E}}_X X_l$

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- **Decomposition of demand elasticity:**

$$\begin{aligned} \mathcal{E} &= \text{relative elasticity} + \text{substitution} \\ &= \text{diagonal matrix} + X \underbrace{\boldsymbol{\mathcal{E}}_X}_{K \times K} X' \end{aligned}$$

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  - Assume constant relative elasticity $\widehat{\mathcal{E}}$ for simplicity, relax in the paper

Questions Revisited

$$\begin{aligned} \mathcal{E} &= \text{relative elasticity} + \text{substitution} \\ &= \hat{\mathcal{E}} I + X \mathcal{E}_X X' \end{aligned}$$

Different questions are about different parts of $\mathcal{E}$

- *How does CalPERS adjust its position in 10-year corporate bonds of Ford and GM when their spread changes?*
  - Asset-specific behavior characterized by the relative elasticity $\hat{\mathcal{E}}$
- *How does CalPERS change its green tilt when the price of greenium increases?*
  - Question about substitution characterized by $\mathcal{E}_X$

Many Potential Mechanisms

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- *Broad categories*: X_i are group dummies say on durations or industries
- *Non-risk motives*: X_i is asset weight in this objective

$$\max_{D} \quad D'(\mu - P) - \frac{\gamma}{2} D'\Sigma D - \frac{\kappa}{2}\left(D'X^{(1)}\right)^2$$
$$\text{such that} \quad D'X^{(2)} \leqslant \Theta$$

  - Binding constraints (leverage), regulatory score (capital ratio), or stakeholders pressure (greenness)

Outside of Finance

Food shopping

- Choose a basket → distinct from discrete choice models
- Items contribute multiple dimensions in multiple quantities → distinct from nest models
  - peanut butter gives you protein and fat
  - peanut butter gives you less protein than chicken breast

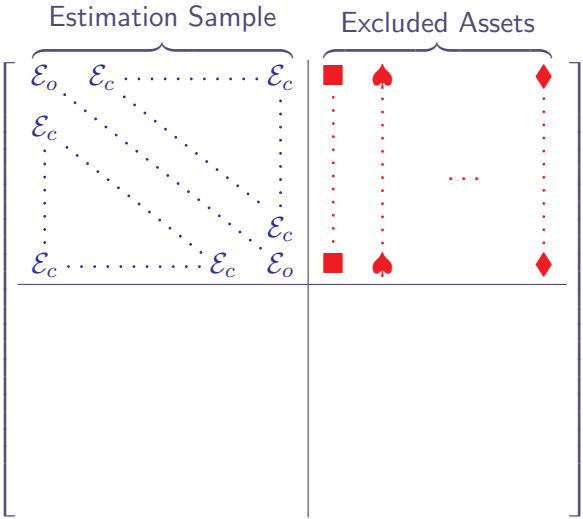
Outside of Finance

Food shopping

- Choose a basket → distinct from discrete choice models
- Items contribute multiple dimensions in multiple quantities → distinct from nest models
  - peanut butter gives you protein and fat
  - peanut butter gives you less protein than chicken breast
- *Homogeneous substitution conditional on observables:* you enjoy all kinds of foods, but want to balance your diet on a few dimensions (e.g. protein, carbs, fat)

Experimental Design, Forming Spreads

- With few close assets: ignore observables & assume full homogeneity
  - Same own- and cross-price elasticity for every pair of assets in $\mathcal{S}$
  - Demand for all assets in $\mathcal{S}$ responds in same way to price of 5-year First Solar bond (outside $\mathcal{S}$)



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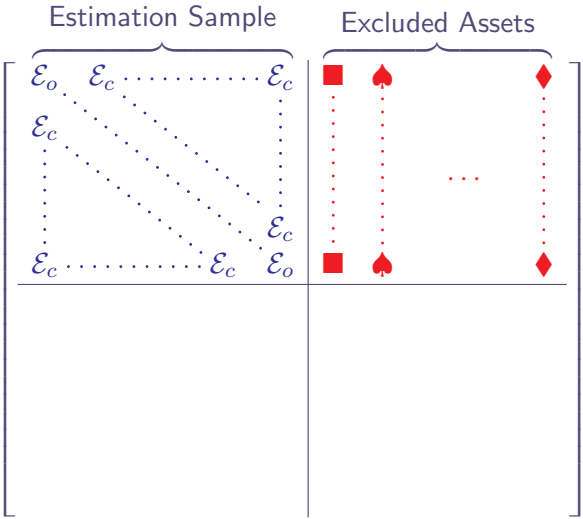

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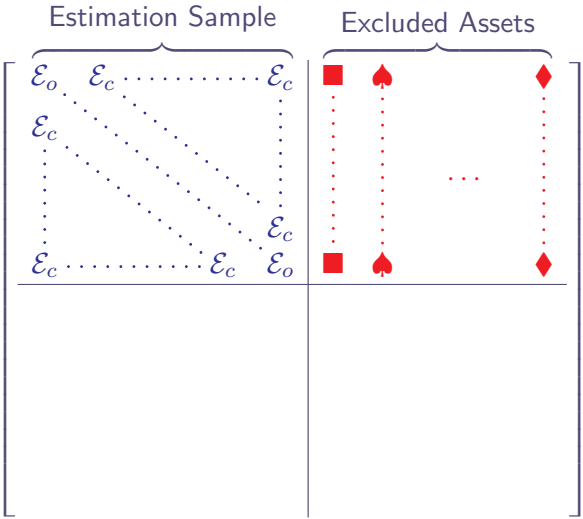
- Diagnostic: balance between treated (high Z_i) and control (low Z_i) on covariance with broad factors



Estimation

Cross-Sectional Identification

- **Data-Generating-Process:** Elasticity matrix $\boldsymbol{\mathcal{E}}$ + *homogeneous substitution conditional on observable* X

$$\Delta \mathbf{D} = \boldsymbol{\mathcal{E}} \Delta \mathbf{P} + \epsilon$$

  - Demand shift ϵ correlated with prices: Ford is more expensive because the new F150 is amazing, change in CalPERS financial health, ...

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- **Proposition 1** Under our assumption, and the *usual exclusion and relevance restrictions*, the IV estimator identifies the **relative elasticity** $\widehat{\mathcal{E}} = \boldsymbol{\mathcal{E}}_{ii} - \boldsymbol{\mathcal{E}}_{ji}$ for $X_i = X_j$

$$\Delta D_i = \widehat{\mathcal{E}} \Delta P_i + \theta' X_i + e_i$$
$$\Delta P_i = \lambda Z_i + \eta' X_i + u_i$$

  with Z_i instrument for prices ($Z_i \perp \epsilon_i | X_i$)
  - E.g.: Fed buys some bonds but not others

Absorbing Substitution

- Key step: coefficient on observables θ absorbs substitution from other assets

$$\begin{aligned}\Delta D_i &= \mathcal{E}_{ii}\Delta P_i + \sum_{j\neq i} X_i'\mathcal{E}_X X_j \Delta P_j + \epsilon_i \\ &= \left(\mathcal{E}_{ii} - X_i'\mathcal{E}_X X_i\right)\Delta P_i + \sum_j X_i'\mathcal{E}_X X_j \Delta P_j + \epsilon_i \\ &= \underbrace{\left(\mathcal{E}_{ii} - X_i'\mathcal{E}_X X_i\right)}_{\text{relative elasticity}}\Delta P_i + X_i' \underbrace{\sum_j \mathcal{E}_X X_j \Delta P_j}_{\text{constant across assets, absorbed in } \theta} + \epsilon_i\end{aligned}$$

- **Relative elasticity:** difference between own-price and cross-price elasticity for assets with same observables
  - How does the relative demand for Ford and GM respond to their relative price?
  - Useful to answer relative Qs and construct relative counterfactuals
  - In large cross-sections with substantial idiosyncratic risk $\approx$ own-price elasticity
  - What GE theorists call the Morishima elasticity, Gabaix Koijen 2025 the micro elasticity

Substitution: The Missing Coefficient Problem

Proposition 2 **Impossible to identify substitution with the cross-section alone**

$$\Delta D_i = \hat{\mathcal{E}} \Delta P_i + X_i' \sum_j \boldsymbol{\mathcal{E}}_X X_j \Delta P_j + \underbrace{\epsilon_i}_{=X_i'\theta_0+\eta_i}$$

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- Coefficient on X_i measures both substitution and shift in demand for observable
  - Does CalPERS reduce its green tilt because of expensive green bonds or weaker environmental priorities?

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- Logit: makes θ a structural parameter ... and cannot capture this substitution

Re-Packaging Substitution

Classic strategy: construct portfolios sorted on observables, and measure their price and demand (= portfolio tilt)

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- Form Fama–MacBeth portfolios:

$$\Delta D_X = (X'X)^{-1}X'\Delta D \qquad \Delta P_X = (X'X)^{-1}X'\Delta P$$

$$\Delta D_{idio,i} = \Delta D_i - X_i'\Delta D_X \qquad \Delta P_{idio,i} = \Delta P_i - X_i'\Delta P_X$$

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- *Decompose between the relative part and substitution between the K portfolios:*

Micro: $$\Delta D_{idio,i} = \widehat{\mathcal{E}}\Delta P_{idio,i}$$

Meso-Macro: $$\Delta D_X = \breve{\mathcal{E}}\Delta P_X$$

where $\breve{\mathcal{E}} = \widehat{\mathcal{E}}\mathbf{I}_K + \mathcal{E}_X X'X$

Estimating Substitution with the Time Series

- **Proposition 3** Regressing portfolio tilts on portfolio prices with **time series instruments** identifies substitution $\mathcal{E}_X$

$$\Delta D_{X,t} = \check{\boldsymbol{\mathcal{E}}} \Delta P_{X,t} + \epsilon_{X,t}$$

  - Effectively only K assets = portfolios
  - E.g. Fed does more or less QE and operation twist over time
  - E.g. granular IVs for different portfolios (Gabaix Koijen 2024)

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- No more separation, even for macro elasticity
  - Is your shock to the "price of stocks" a parallel shift, or concentrated in small stocks? tech stocks?

Summary

Homogeneous substitution conditional on observables X:

$$\begin{aligned} \mathcal{E} &= \text{relative elasticity} + \text{substitution} \\ &= \quad \widehat{\mathcal{E}} I \quad + \quad X \boldsymbol{\mathcal{E}}_X X' \end{aligned}$$

Consistent with many motives: risk, constraints, non-pecuniary preferences, irrational, ...

Identification:

- Relative elasticity: compare similar assets = cross-sectional IV controlling for X
- Substitution: demand for portfolios based on X = time-series portfolio level instruments
- Even purely cross-sectional questions rely on both relative elasticity AND substitution

Comparison to Other Models

What About Logit?

- Koijen Yogo (2019), Koijen Richmond Yogo (2024):
  - $\exists$ factor models where volatility and expected returns depend non-linearly on prices which yield asset demand in the logit form
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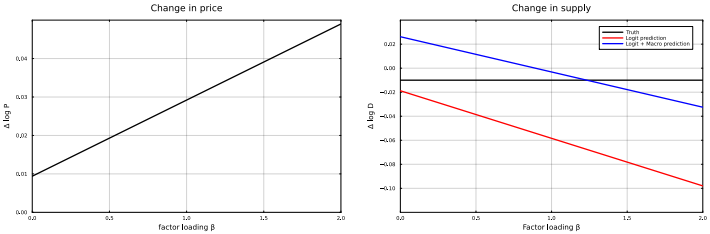
- Logit satisfies our assumption, and its parameter can be robustly interpreted as relative elasticity
- Logit strongly restricts substitution: **an arbitrary factor model is not equivalent to logit**
  - Logit: when the price of any bond $\uparrow$, CalPERS replaces it proportionally to its existing portfolio, substitution $\omega\omega'$
  - Factor model: CalPERS replaces it disproportionately with bonds loading on similar factors, substitution $\beta\mathcal{E}_X\beta'$

Understanding the Difference



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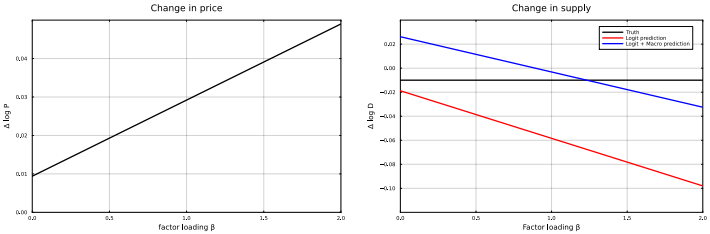

- Recall equal amount of Δ purchase $\Rightarrow$ higher Δ price for higher β asset
- The **missing coefficients** problem in asset demand under a factor model
  - $\theta(\{P_j\}) = \sum_j \mathcal{E}_X X_j \Delta P_j$ which hits X_i depends on counterfactual price vector
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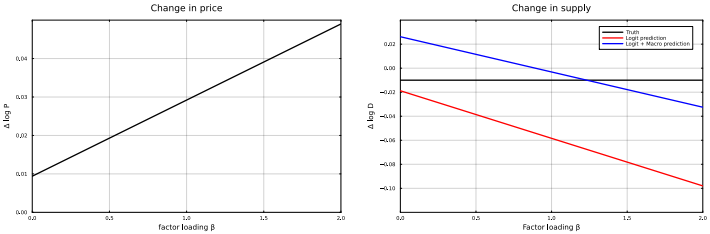

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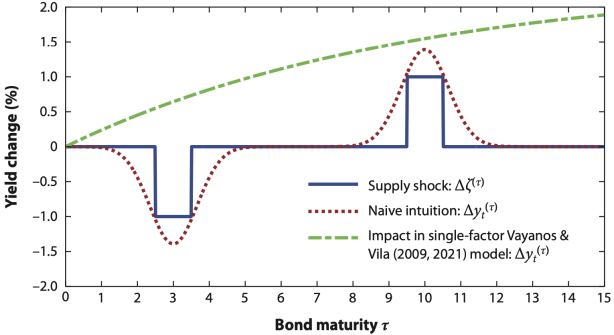
  - Taking this into account $\Rightarrow$ correct P-D relationship
- In both Logit and Micro-Macro, treat θ as an exogenous coefficient for X_i
  - In Logit, let D_0 as the outside asset weight, then ($\widehat{\mathcal{E}} < 0$)

$$D_i(\{P_j\}) = D_0(\{P_j\}) \cdot e^{\widehat{\mathcal{E}} \cdot \ln P_i + X_i' \theta} \Leftrightarrow \ln D_i(\{P_j\}) = \ln D_0(\{P_j\}) + \widehat{\mathcal{E}} \ln P_i + X_i' \theta$$

  - $D_0(\{P_j\})$ shifts demand uniformly across all other assets with different X's
  - Only solves **missing intercept** problem, like in Micro-Macro (identical individual assets)

Group-based substitution vs Factor Models

- Could use observables as continuous variables (factor loading) or categories (nested models)
- Nested logit (Fang 2023, Koijen Yogo 2024): symmetric groups based on values of observables → can use the cross-section of groups to estimate substitution
  - Predict strong local effect and diffuse effect across all other groups
  - Sharply different from factor model with exposure depending on observable (see Cochrane 2008, Vayanos Vila 2021) conclusion



Price Impact in Corporate Bonds

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- Same structure: $\boldsymbol{\mathcal{M}} = \widehat{\mathcal{M}}\mathbf{I} + X\boldsymbol{\mathcal{M}}_X X'$, and same estimation
  - Cross-section reveals relative multiplier $\widehat{\mathcal{M}}$
  - Time series of portfolios reveals spillovers $\boldsymbol{\mathcal{M}}_X$

Price Impact for Corporate Bonds

- U.S. corporate bonds, 2010–2024 (from WRDS Bond Returns) (following Chaudhary Fu Li 2024)
- *Source of demand shocks:* flow-induced trading (Coval Stafford 2007, Lou 2012, Chaudhry 2025)

$$Z_{it} = \sum_k \frac{A_{k,t-1}\widetilde{w}_{i,k,t-1}}{P_{i,t-1}S_{i,t-1}} f_{kt}$$

- *Exogeneity condition:* $Z_{it} \perp \epsilon_{it} | X_{it}$
  - example threat to identification: another investor type (e.g., insurance companies) buying the same bonds held by specific mutual funds

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Do treated & control bonds with the same values of observables comove the same way with broad portfolios?

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1. At each date t, form a long-short portfolio based on treatment status
   - $Z_{idio,it}$: residual of instrument regressed on duration $\times$ date and credit rating $\times$ date fixed effects
2. Compute the β of the long-short return on broad indices in a window around t (here: 2y)
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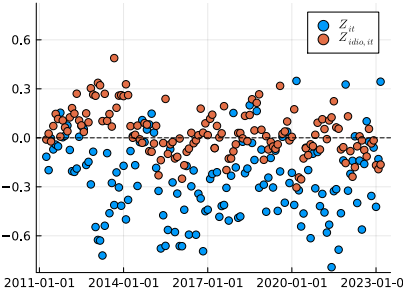
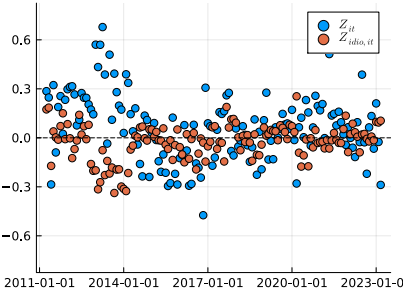
- Similar diagnostic for constant relative elasticity: balance on idiosyncratic volatility

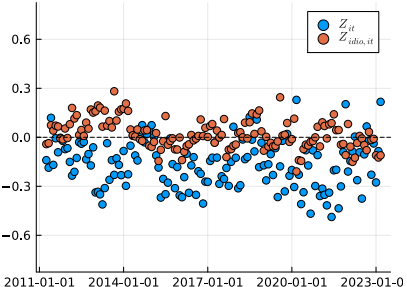
A. Corporate Bond Index



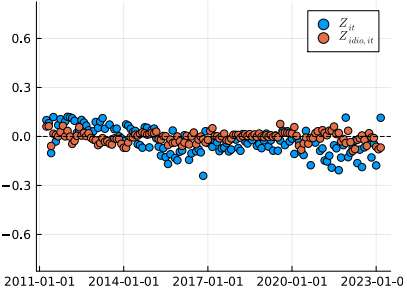
B. High−Low Credit Rating



C. Long−Short Term Bonds



D. Stock Index



Relative Multiplier

Relative multiplier $\widehat{\mathcal{M}}$

	Return $\Delta P_{it}/P_{i,t-1}$				
	(1)	(2)	(3)	(4)	(5)
Demand shock:					
Z_{it}	2.080***	0.492***	**0.055**		
	(0.387)	(0.128)	(0.084)		
$Z_{idio,it}$				**0.055**	**0.055**
				(0.084)	(0.087)
Date Fixed Effects		Yes	Yes	Yes	Yes
Duration × Date Fixed Effects			Yes	Yes	
Credit Risk × Date Fixed Effects			Yes	Yes	
N	1,041,985	1,041,985	1,041,985	1,041,985	1,041,985
R^2	0.018	0.293	0.464	0.464	0.293

→ A change in demand for one bond vs. another with the same duration and credit rating does not move their relative spread

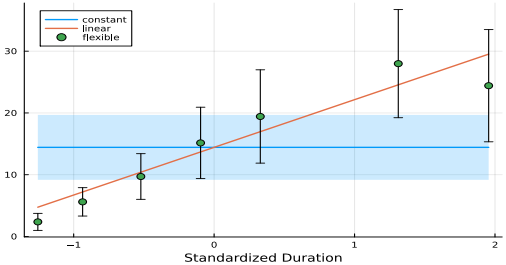
Spillovers: Cross-Multipliers

	Return				
	$\frac{\Delta P_{agg,t}}{P_{agg,t-1}}$		$\frac{\Delta P_{DUR,t}}{P_{DUR,t-1}}$	$\frac{\Delta P_{PD,t}}{P_{PD,t-1}}$	$\frac{\Delta P_{it}}{P_{i,t-1}}$
	(1)	(2)	(3)	(4)	(5)
$Z_{agg,t}$	15.032***	**14.430*** **	**7.773*** **	**3.764** **	14.430***
	(2.393)	(2.707)	(1.803)	(1.392)	(2.689)
$Z_{DUR,t}$		**0.425**	**4.531**	**-5.789**	0.425
		(6.713)	(4.293)	(3.948)	(6.646)
$Z_{PD,t}$		**1.127**	**-0.868**	**3.383** **	1.127
		(1.866)	(1.404)	(1.025)	(1.854)
$Z_{idio,it}$					0.055
					(0.087)
Duration DUR_{it}					0.001
					(0.001)
N	168	168	168	168	1,041,985
R^2	0.342	0.343	0.173	0.384	0.101

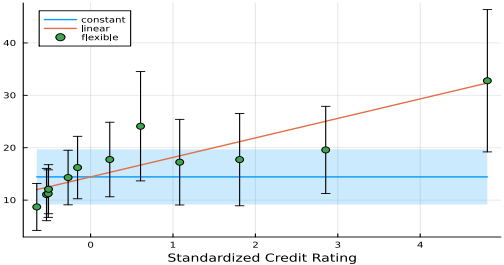
- Spillovers are not uniform: line up with duration and credit rating
- Fed buys 1% of supply of all corporate bonds → long-duration bonds rise ∼22%, short-duration ∼7%
- Fed buys 1% long, sells 1% short duration (operation twist) → long-duration +5%, short-duration −4%

Unpacking the Spillovers (Overidentification)

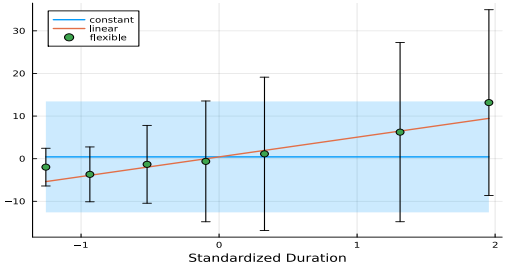
A. Response to Z_{agg} by Duration



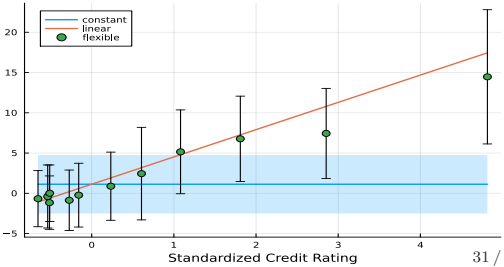
B. Response to Z_{agg} by Credit



C. Response to Z_{DUR} by Duration



D. Response to Z_{PD} by Credit



Conclusion

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- To draw causal inference about demand elasticity, need flexible substitution = portfolio management
  - A simple assumption: **homogeneous substitution conditional on observables**
    - CalPERS substitutes based on duration and greenness
  - (Standard) source of exogenous variation
    - Fed randomly buys more of some bonds than others, Fed surprisingly engages in QE
- Relative elasticity for similar assets: cross-sectional IV
  - Ford vs GM?
- Substitution = demand for portfolios: time-series IV
  - Green vs brown? Aggregate price?
- Need both dimensions even for cross-sectional questions
- Standard structural models of demand rule out most factor-style substitution

Why Causal Inference in Asset Pricing?

- Causal inference particularly valuable when:
  - existing theories are far from the data
  - it is challenging to understand all sources of variations simultaneously
- First step towards better economic theory

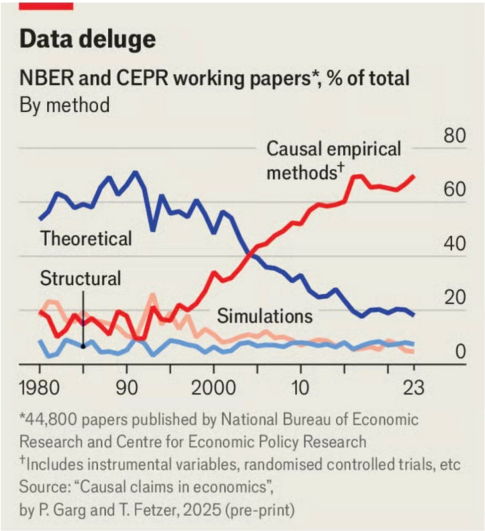

CHART: THE ECONOMIST